Q101 Ultra-Low Power GPS Disciplined Oscillator

Key Features

- Ultra-low power consumption <125 mW
- 1PPS input and 1PPS output for GPS synchronization
- Discipline to 1 ns RMS in phase and $<1x10^{-12}$ in frequency
- 1 Second continuous phase measurement and report system, resolution ≤ 1 ns
- ToD (Time of Day)
- RS232 digital interface
- User accessible data memory, optional
- 5,10 MHz, 16.384 MHz, HCMOS output
- <±1.5us, 24 hour, +25C



Applications

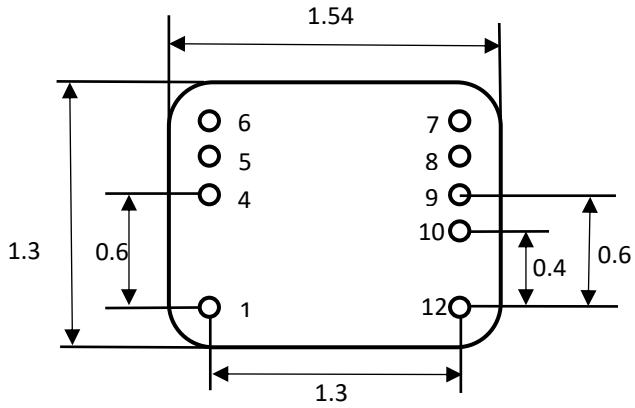
- Low Orbit Satellite
- UVA
- Avionics
- Battery Powered Communication Systems
- Under Water Autonomous Vehicle

Mechanical Specifications

- **Pin Assignment (*Consulting factory for other pin compatible footprint*)**

Pin #	Assignment
1	NC
4	Oven Ready
5	Tx
6	Rx
7	+3.3 V
8	GND
9	1PPS input
10	1PPS output
12	RF output

Bottom view



- **Mechanical Dimension** 1.54” x 1.34” x 0.60”

TimeCubic Proprietary

Electrical & Environmental Specifications (25 °C, +3.3 VDC input unless otherwise specified)

PARAMETERS	Unit	Min	Typ.	Max	Note	Comments
RF Output						
• Frequency	MHz		16.384			
• Amplitude	V		2.6			CMOS
• Load	MΩ//10 pf		10			
• R/F Time	ns			10		
1PPS Time Output						
• Pulse Width	µs		20			
• Amplitude	V		3.3			CMOS
• R/F Time	ns			10		
• Load	MΩ//10 pf		10			
1PPS Time Input						
• Format			Rising edge			
• Amplitude	V		3.3			CMOS
• Impedance	MΩ		10			
Digital Communication						
• Protocol			RS232			
• Logic Level	V		3.3			CMOS
• Baud Rate			57600			
Power Input						
• DC Input	V		3.3			
• Power Consumption	mW			95		-10 °C to +70 °C
	mW			125		-40 °C to +85 °C
• Warm up Power	mW			550		*Factory configurable*
• Warm up Time	Minute	1		4		*Factory configurable for fast warm up*
Environmental						
• Operating Temperature & Stability	ppb(p-p)			1.5		-10 °C to +70 °C
	ppb(p-p)			2.0		-40 °C to +85 °C
• Mechanical Shock						Mil-STD-202, >100 G, 11 ms
• Vibration						Mil-STD-202, 5 G up to 2 kHz
Stability						
• Acceleration Sensitivity	/g		$\pm 1 \times 10^{-9}$			Worst direction
• Aging (after 30days)	/day			$\pm 2 \times 10^{-10}$		$\pm 0.5 \times 10^{-8}$/first year
• Frequency Control	ppm			±0.7		digital

Holdover

- <±1.5us for 24 hour, ±2C, 7 day on, 1 day discipline

Consult factory for other waveform, Consulting factory for temperature range, stability, warmup power ad warmup time.

TimeCubic Proprietary

Order Information

Temperature Range

A	-10C/+70C
B	-45C/+85C

Output

Code	Spec.
C	CMOS

Q100 10 A 20 C

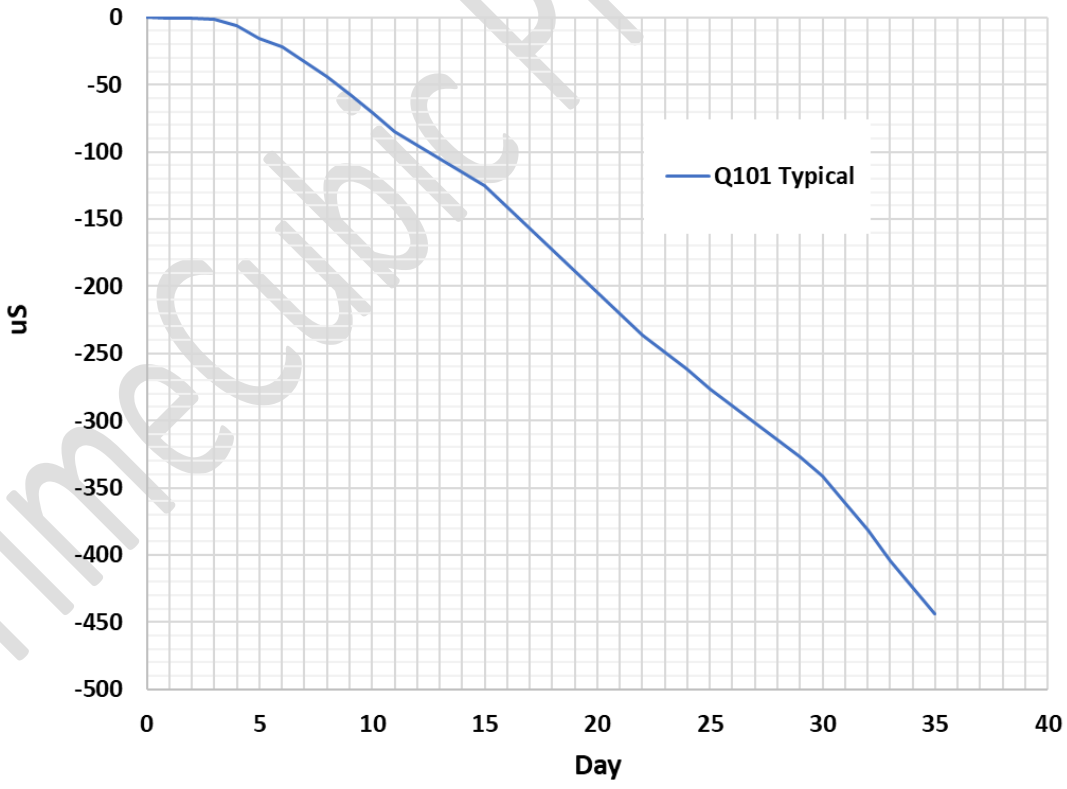
Frequency (MHz)

Code	Spec.
10	10

Aging/Day

Code	Spec.
20	$\pm 2 \times 10^{-10}$

Typical holdover performance, +25C ± 2C, active correction enabled



TimeCubic Proprietary